Rademacher and Gaussian Complexities: Risk Bounds and Structural Results

Paper Presentation

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2. Risk Bounds
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   - Kernel Methods
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Introduction

Normalized Risk Bound

Consider the binary classification setting. Let D is a distribution over $\mathcal{X} \times \{\pm 1\}$. Let F be a hypothesis class of VC-dimension d. Consider ℓ the $0-1$ loss i.e. $\ell(y, y') = \mathbf{1}(y \neq y')$.

Denote $\hat{\mathbb{E}}_n[\ell(Y, f(X))] = \frac{1}{n} \sum_{i=1}^{n} \ell(Y_i, f(X_i))$, where $(X_i, Y_i)_{i=1}^n \sim D^n$.

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Then, with probability at least $1-\delta$, for every $f \in F$,

$$\frac{\mathbb{E}[\ell(Y, f(X))] - \hat{\mathbb{E}}_n[\ell(Y, f(X))]}{\hat{\mathbb{E}}_n[\ell(Y, f(X))]} \leqslant 2\sqrt{\frac{d \ln(2en/d) + \ln(4/\delta)}{n}} := c(d, n).$$ [1]

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This implies

$$\mathbb{E}[\ell(Y, f(X)) \leqslant \hat{\mathbb{E}}_n[\ell(Y, f(X))] + \frac{c^2(d, n)}{2}\left(1 + \sqrt{1 + \frac{4\hat{\mathbb{E}}_n[\ell(Y, f(X))]}{c^2(d, n)}}\right).$$

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Model Selection by Normalized Risk Bound

One minimizes RHS of (2) over classes of interest with $\hat{\mathbb{E}}_n[\ell(Y, f(X))]$ replaced by the train error.

In such methods, the chosen complexity of a class (VC dimension) is independent of data.

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Problems

- None of such methods is universally effective [2].
- The bounds are loose (requires the sample size n be large enough to be practical, since otherwise, $RHS \gg 1$).

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Complexity penalties depending on train data are developed.

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Data-Dependent Complexities

Empirical Rademacher complexity

$$\hat{R}_n(F) = \mathbb{E}\left[\sup_{f \in F}\left|\frac{2}{n}\sum_{i=1}^{n}\sigma_i f(X_i)\right| \,\middle|\, X_1, \ldots, X_n\right],$$

where $\sigma_1, \ldots, \sigma_n$ are i.i.d Rademacher random variables.

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Notes: different conventions

- The factor is 2
- Absolute value is taken by default.

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Empirical Gaussian complexity

$$\hat{G}_n(F) = \mathbb{E}\left[\sup_{f\in F}\left|\frac{2}{n}\sum_{i=1}^{n}g_i f(X_i)\right| \,\middle|\, X_1,\ldots,X_n\right],$$

where $g_1,\ldots,g_n$ are i.i.d standard Gaussian random variables.

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where $g_1,\ldots,g_n$ are i.i.d standard Gaussian random variables.

Expected complexities: $R_n(F) = \mathbb{E}[\hat{R}_n(F)]$ and $G_n(F) = \mathbb{E}[\hat{G}_n(F)]$.

Model Selection through Data-Dependent Complexities

Other problems towards practice

1. Bounds through relaxed loss functions that are friendly to optimization algorithms are desired.
2. Computing the complexities is optimizing over F, which is intractable with large F of interest.
   - Paper's solution: estimating by expressing F as a combination of basic classes.

Risk Bounds

Dominating, Lipschitz Loss Relaxation

We recall the usual setting

1. Let D be a distribution over $\mathcal{X} \times \{\pm 1\}$, $(X, Y) \sim D$ and $(X_i, Y_i)_{i=1}^n \sim D^n$.
2. Let F be a class of soft classifiers of the form $f : \mathcal{X} \to \mathbb{R}$. To each $f \in F$ corresponds a hard classifier $\mathrm{sgn} \circ f$ i.e. $\mathbb{P}(Y \neq \mathrm{sgn} \circ f(X)) \leqslant \mathbb{P}(Yf(X) \leqslant 0)$.

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Theorem 2

Suppose that every $f \in F$ has a finite supremum. Let $\phi : \mathbb{R} \to [0, 1]$ be L-Lipschitz and dominate the $0 - 1$ loss i.e. $\forall \alpha \in \mathbb{R}, \phi(\alpha) \geqslant \mathbf{1}(\alpha \leqslant 0)$. Then, with probability at least $1 - \delta$, for every $f \in F$,

$$\mathbb{P}(Yf(X) \leqslant 0) \leqslant \hat{\mathbb{E}}_n[\phi(Yf(X))] + 2LR_n(F) + \sqrt{\frac{\ln(2/\delta)}{2n}}$$

Dominating Loss Relaxation

The setting is decision-theoretic.

1. Let D be a distribution over $\mathcal{X} \times \mathcal{Y}$, $(X, Y) \sim D$ and $(X_i, Y_i)_{i=1}^n \sim D^n$.
2. Let $\mathcal{A}$ be the action space and F be a class of decision functions of the form $f : \mathcal{X} \to \mathcal{A}$ (commonly $\mathcal{A} = \mathcal{Y}$).

Theorem 3

Let $\ell, \phi : \mathcal{Y} \times \mathcal{A} \to [0, 1]$ be loss functions such that ϕ dominates ℓ i.e. $\forall y \in \mathcal{Y}, a \in \mathcal{A}, \phi(y, a) \geqslant \ell(y, a)$. Then, with probability at least $1 - \delta$, for every $f \in F$,

$$\mathbb{E}[\ell(Y, f(X))] \leqslant \hat{\mathbb{E}}_n[\phi(Y, f(X))] + R_n(\tilde{\phi} \circ F) + \sqrt{\frac{8 \ln(2/\delta)}{n}},$$

where $\tilde{\phi} \circ F = \{(x, y) \mapsto \phi(y, f(x)) - \phi(y, 0) \mid f \in F\}$

Estimating Complexities

Estimating Complexities

Neural Networks

Neural Networks

Theorem 4

Consider an L-Lipschitz activation $\sigma : \mathbb{R} \to [-1, 1]$ being monotonic and smooth. Define a class of two-layer neural networks

$$F = \left\{ x \mapsto \sum_i w_i \sigma(v_i \cdot x) \ : \ \|w\|_1 \leqslant 1 \text{ and } \forall i, \|v_i\|_1 \leqslant B \right\}.$$

Then for any fixed sample $x_1, ..., x_n \in \mathbb{R}^k$, we have

$$\hat{G}_n(F) := \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n g_i f(x_i) \right] \leqslant \frac{LB(8 \ln k)^{1/2}}{n} \max_{j,j'} \sqrt{\sum_{i=1}^n (x_{ij} - x_{ij'})^2}$$

Neural Networks

Proof. Let $H = \{x \mapsto v \cdot x : \|v\|_1 \leqslant 1\}$, then $F = \mathrm{conv}(\sigma \circ BH)$. We have (with admitted lemmas)

$$\hat{G}_n(F) = \hat{G}_n(\sigma \circ BH)$$

invariance under convex hull

Neural Networks

Proof. Let $H = \{x \mapsto v \cdot x : \|v\|_1 \leqslant 1\}$, then $F = \mathrm{conv}(\sigma \circ BH)$. We have (with admitted lemmas)

$$\hat{G}_n(F) = \hat{G}_n(\sigma \circ BH) \leqslant L\hat{G}_n(BH)$$

scaling with Lipschitz composition

Neural Networks

Proof. Let $H = \{x \mapsto v \cdot x : \|v\|_1 \leqslant 1\}$, then $F = \mathrm{conv}(\sigma \circ BH)$. We have (with admitted lemmas)

$$\hat{G}_n(F) = \hat{G}_n(\sigma \circ BH) \leqslant L\hat{G}_n(BH) \leqslant LB \cdot \hat{G}_n(H)$$

homogeneity

Neural Networks

Proof. Let $H = \{x \mapsto v \cdot x : \|v\|_1 \leqslant 1\}$, then $F = \mathrm{conv}(\sigma \circ BH)$. We have (with admitted lemmas)

$$\begin{aligned}\hat{G}_n(F) &= \hat{G}_n(\sigma \circ BH) \leqslant L\hat{G}_n(BH) \leqslant LB \cdot \hat{G}_n(H) \\ &= LB \cdot \mathbb{E}\left[\max_{1 \leqslant j \leqslant k} \frac{2}{n} \sum_{i=1}^{n} g_i x_{ij}\right]\end{aligned}$$

H is symmetric

Neural Networks

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let $Z_j = \sum_{i=1}^{n} g_i x_{ij}$

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&\leqslant \frac{2}{n} LB \cdot \sqrt{2 \ln k} \max_{1 \leqslant j, j' \leqslant k} \sqrt{\mathbb{E}\left[(Z_j - Z_{j'})^2\right]}
\end{aligned}$$

Ledoux-Talagrand 1991, followed from Slepian's Lemma

Neural Networks

Proof. Let $H = \{x \mapsto v \cdot x : \|v\|_1 \leqslant 1\}$, then $F = \mathrm{conv}(\sigma \circ BH)$. We have (with admitted lemmas)

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\end{aligned}$$

Note. When $\|w\|_1 \leqslant W$, the bound simply scales by W.

Estimating Complexities

Kernel Methods

Kernel Methods

A kernel $k : X \times X \to \mathbb{R}$ on a compact space X is a continuous symmetric positive semidefinite function. Kernel methods use expansions of the form

$$f(x) = \sum_{i=1}^{n} \alpha_i k(X_i, x).$$

Fix $\gamma > 0$, and define the margin cost

$$\phi(\alpha) = \begin{cases} 1, & \alpha \leqslant 0, \\ 1 - \alpha/\gamma, & 0 < \alpha \leqslant \gamma, \\ 0, & \alpha > \gamma. \end{cases}$$

So, as in the previous sections, the idea is:

$$\text{true classification error} \leqslant \text{empirical margin loss} + \text{complexity term}.$$

Kernel methods

Theorem 5

Fix $B, \gamma > 0$, and let $k : X \times X \to \mathbb{R}$ be a kernel with

$$\sup_{x \in X} |k(x, x)| < \infty.$$

Suppose $(X_1, Y_1), \ldots, (X_n, Y_n)$ are i.i.d. from P on $X \times \{\pm 1\}$. Then with probability at least $1 - \delta$, every function

$$f(x) = \sum_{i=1}^{n} \alpha_i k(X_i, x)$$

with $\sum_{i,j} \alpha_i \alpha_j k(X_i, X_j) \leqslant B^2$ satisfying

$$P(Yf(X) \leqslant 0) \leqslant \hat{\mathbb{E}}_n \phi(Yf(X)) + \frac{4B}{\gamma n} \sqrt{\sum_{i=1}^{n} k(X_i, X_i)} + \left(\frac{8}{\gamma} + 1\right) \sqrt{\frac{\ln(4/\delta)}{2n}}.$$

Kernel methods

Proof. Apply the general margin bound from Section 2 to the class

$$F_B = \{x \mapsto \langle w, \Phi(x)\rangle_{\mathcal{H}} : \ \|w\|_{\mathcal{H}} \leqslant B\}$$

with the margin loss ϕ, which is $1/\gamma$-Lipschitz.

This gives

$$P(Yf(X) \leqslant 0) \leqslant \hat{\mathbb{E}}_n \phi(Yf(X)) + \frac{2}{\gamma} R_n(F_B) + \text{confidence term.}$$

Kernel methods

By definition,

$$
\begin{aligned}
\hat{R}_n(F_B) &= \frac{2}{n}\mathbb{E}_g\left[\sup_{\|w\|_{\mathcal{H}}\leqslant B}\left\langle w, \sum_{i=1}^n \sigma_i\Phi(X_i)\right\rangle\right] \\
&\leqslant \frac{2B}{n}\,\mathbb{E}_\sigma\left\|\sum_{i=1}^n \sigma_i\Phi(X_i)\right\|_{\mathcal{H}} && \text{(Cauchy-Schwarz)} \\
&\leqslant\leqslant \frac{2B}{n}\left(\mathbb{E}_\sigma\left\|\sum_{i=1}^n \sigma_i\Phi(X_i)\right\|_{\mathcal{H}}^2\right)^{1/2} && \text{(Jensen's)} \\
&= \frac{2B}{n}\left(\mathbb{E}_\sigma\left[\sum_{i,j=1}^n \sigma_i\sigma_j\langle\Phi(X_i),\Phi(X_j)\rangle_{\mathcal{H}}\right]\right)^{1/2} \\
&= \frac{2B}{n}\left(\mathbb{E}_\sigma\left[\sum_{i,j=1}^n \sigma_i\sigma_j k(X_i,X_j)\right]\right)^{1/2}
\end{aligned}
$$

Kernel methods

Then, taking expectation, we have

$$\mathbb{E}_g \left\| \sum_{i=1}^n \sigma_i \Phi(X_i) \right\|_{\mathcal{H}}^2 = \sum_{i=1}^n k(X_i, X_i) \qquad \text{(independence of Gaussians: } \mathbb{E}[\sigma_i \sigma_j] = 0 \text{ for } i \neq j\text{)}.$$

Therefore

$$\hat{R}_n(F_B) \leqslant \frac{2B}{n} \sqrt{\sum_{i=1}^n k(X_i, X_i)}.$$

Substitute this into the margin bound:

$$P(Yf(X) \leqslant 0) \leqslant \hat{\mathbb{E}}_n \phi(Yf(X)) + \frac{4B}{\gamma n} \sqrt{\sum_{i=1}^n k(X_i, X_i)} + \left(\frac{8}{\gamma} + 1 \right) \sqrt{\frac{\ln(4/\delta)}{2n}}.$$

Estimating Complexities

Decision Trees

Decision Trees

Theorem 6

Let D be a probability distribution on $\mathcal{X} \times \{\pm 1\}$, $(X, Y) \sim D$. Let H be a set of $\{0, 1\}$-functions on $\mathcal{X}$. Let T be the class of decision trees of depth no more than d, with decision functions from H (to be clear later).

For a training sample and a tree $t \in T$, let $\tilde{P}_n(\ell)$ denote the proportion of all training examples which reach leaf ℓ and are correctly classified.

Then with probability at least $1 - \delta$, there exists a constant c such that every $t \in T$ with L leaves satisfies

$$\mathbb{P}(Y \neq t(X)) \leqslant \hat{\mathbb{P}}_n(Y \neq t(X)) + \sum_{\ell} \min\big(\tilde{P}_n(\ell),\, cd\, G_n(H)\big) + \sqrt{\frac{c \ln(L/\delta)}{2n}}.$$

Decision Trees

For a tree t of depth d, the indicator of whether an input reaches a leaf ℓ is a conjunction of at most $d_\ell \leqslant d$ node decision functions

$$\mathbf{1}_{\{x \to \ell\}} = \bigwedge_{i=1}^{d_\ell} h_{\ell,i}(x), \qquad h_{\ell,i} \in H.$$

Let $\sigma_\ell \in \{\pm 1\}$ be the label assigned at leaf ℓ.

For every $x \in \mathcal{X}$, $t(x) = \sigma_{\ell *}$ for some ℓ^*.

Hence, for *any* $w_\ell > 0$ such that $\sum_\ell w_\ell = 1$, the function f defined by

$$f(x) = \sum_\ell w_\ell \sigma_\ell \bigwedge_{i=1}^{d_\ell} h_{\ell,i}(x),$$

is a soft classifier of t. Let F be the class of such functions.

We will bound $\mathbb{P}(Yf(X) \leqslant 0)$.

Decision Trees

Define $\phi_K : [0,1] \to [0,1]$ such that $\phi_K(\alpha) = \max(0, 1 - K\alpha)$, then ϕ_K is K-Lipschitz and dominates the $0-1$ loss.

By Theorem 6, for each $K \in \mathbb{N}$ (chosen before f), with probability $1 - \delta_K$

$$\mathbb{P}(Yf(X) \leqslant 0) \leqslant \hat{\mathbb{E}}_n \phi_K(Yf(X)) + 2KR_n(F) + \sqrt{\frac{\ln(2/\delta_K)}{2n}}.$$

Choose $\delta_K = \frac{6\delta}{\pi^2 K^2}$ (so that $\sum_{K=1}^{\infty} \delta_K = \delta$). With probability at least $1 - \delta$, all inequalities of the following form hold,

$$\mathbb{P}(Yf(X) \leqslant 0) \leqslant \hat{\mathbb{E}}_n[\phi_K(Yf(X))] + 2KR_n(F) + \sqrt{\frac{\ln(\pi^2 K/3\delta)}{2n}}.$$

Now we can choose K depending on our tree.

Decision Trees

$$\hat{\mathbb{E}}_n[\phi_K(Yf(X))] + 2KR_n(F) = \hat{\mathbb{P}}_n(Yf(X) \leqslant 0) + \sum_{\ell} \tilde{P}_n(\ell)\phi_K(w_\ell) + 2KR_n(F)$$

penalize misclassifications by true loss, penalize correct classifications by low confidence

Decision Trees

$$\begin{aligned}\hat{\mathbb{E}}_n[\phi_K(Yf(X))] + 2KR_n(F) &= \hat{\mathbb{P}}_n(Yf(X) \leqslant 0) + \sum_{\ell} \tilde{P}_n(\ell)\phi_K(w_\ell) + 2KR_n(F) \\ &= \hat{\mathbb{P}}_n(Y \neq t(X)) + \sum_{\ell} \tilde{P}_n(\ell) \max(0, 1 - Kw_\ell) + 2KR_n(F)\end{aligned}$$

Decision Trees

$$\hat{\mathbb{E}}_n[\phi_K(Yf(X))] + 2KR_n(F) = \hat{\mathbb{P}}_n(Yf(X) \leqslant 0) + \sum_{\ell} \tilde{P}_n(\ell)\phi_K(w_\ell) + 2KR_n(F)$$
$$= \hat{\mathbb{P}}_n(Y \neq t(X)) + \sum_{\ell} \tilde{P}_n(\ell) \max(0, 1 - Kw_\ell) + 2KR_n(F)$$

Note the last equality is true even when some $w_\ell = 0$, since $\mathrm{sgn}(0) = 1$. Let x such that $f(x) = 0$

- if $t(x) \neq y$, then $yf(x) \leqslant 0$ is *true*
- if $t(x) = y$, then we are penalized by "no confidence"

Decision Trees

Now let $K = \left|\{\ell : \tilde{P}_n(\ell) > 2R_n(F)\}\right| \leqslant L$, $w_\ell = 0$ if $\tilde{P}_n(\ell) \leqslant 2R_n(F)$ and $w_\ell = \frac{1}{K}$, otherwise.

$$\begin{aligned}\sum_\ell \tilde{P}_n(\ell)\max(0, 1 - Kw_\ell) + 2KR_n(F) &= \sum_\ell \tilde{P}_n(\ell)\cdot \mathbf{1}(\tilde{P}_n(\ell) \leqslant 2R_n(F)) \\ &\quad + 2R_n(F)\sum_\ell \mathbf{1}(\tilde{P}_n(\ell) > 2R_n(F)) \\ &= \sum_\ell \min(\tilde{P}_n(\ell), 2R_n(F))\end{aligned}$$

Decision Trees

Let $g : \{\pm 1\}^d \to \{\pm 1\}$ be the boolean AND function. We express the class F as

$$F \subset \operatorname{conv}\left(\pm \frac{1}{2}\left(g(K^d) + 1\right)\right)$$

where $K = \{k : \mathcal{X} \to \{\pm 1\} : k = 2h - 1, h \in H\}$ (to satisfy the boolean lemma).

Then

$$R_n(F) \leqslant R_n\left(\frac{1}{2}\left(g(K^d) + 1\right)\right)$$

$$\forall F, R_n(F) = R_n(\operatorname{conv}(F)) == R_n(F \cup -F)$$

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Then

$$R_n(F) \leqslant R_n\left(\frac{1}{2}\left(g(K^d) + 1\right)\right) \leqslant \sqrt{\frac{\pi}{2}}\, G_n\left(\frac{1}{2}\left(g(K^d) + 1\right)\right)$$

conversion: $R_n(F) \leqslant \sqrt{\pi/2}\, G_n(F)$

Decision Trees

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Then

$$\begin{aligned} R_n(F) &\leqslant R_n\left(\frac{1}{2}\left(g(K^d) + 1\right)\right) \leqslant \sqrt{\frac{\pi}{2}} G_n\left(\frac{1}{2}\left(g(K^d) + 1\right)\right) \\ &\leqslant \sqrt{\frac{\pi}{2}}\left(\frac{1}{2} G_n(g(K^d)) + \frac{1}{2\sqrt{n}}\right) \end{aligned}$$

translation and scaling: $G_n(cF + h) \leqslant |c| G_n(F) + \|h\|_\infty / \sqrt{n}$

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where $K = \{k : \mathcal{X} \to \{\pm 1\} : k = 2h-1, h \in H\}$ (to satisfy the boolean lemma).

Then

$$\begin{aligned} R_n(F) &\leqslant R_n\left(\frac{1}{2}\left(g(K^d)+1\right)\right) \leqslant \sqrt{\frac{\pi}{2}}\, G_n\left(\frac{1}{2}\left(g(K^d)+1\right)\right) \\ &\leqslant \sqrt{\frac{\pi}{2}}\left(\frac{1}{2}G_n(g(K^d)) + \frac{1}{2\sqrt{n}}\right) \leqslant \sqrt{\frac{\pi}{2}}\left(dG_n(K) + \frac{1}{2\sqrt{n}}\right) \end{aligned}$$

lemma for boolean combination: $G_n(g(K^d)) \leqslant 2dG_n(K)$

Decision Trees

Let $g : \{\pm 1\}^d \to \{\pm 1\}$ be the boolean AND function. We express the class F as

$$F \subset \operatorname{conv}\left(\pm\frac{1}{2}\left(g(K^d)+1\right)\right)$$

where $K = \{k : \mathcal{X} \to \{\pm 1\} : k = 2h-1, h \in H\}$ (to satisfy the boolean lemma).

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$$\begin{aligned} R_n(F) &\leqslant R_n\left(\frac{1}{2}\left(g(K^d)+1\right)\right) \leqslant \sqrt{\frac{\pi}{2}}\, G_n\left(\frac{1}{2}\left(g(K^d)+1\right)\right) \\ &\leqslant \sqrt{\frac{\pi}{2}}\left(\frac{1}{2}G_n(g(K^d)) + \frac{1}{2\sqrt{n}}\right) \leqslant \sqrt{\frac{\pi}{2}}\left(dG_n(K) + \frac{1}{2\sqrt{n}}\right) \\ &\leqslant \sqrt{\frac{\pi}{2}}\left(2dG_n(H) + \frac{2d+1}{2\sqrt{n}}\right) \end{aligned}$$

go back to H

Decision Trees

Let $g : \{\pm 1\}^d \to \{\pm 1\}$ be the boolean AND function. We express the class F as

$$F \subset \operatorname{conv}\left(\pm\frac{1}{2}\left(g(K^d)+1\right)\right)$$

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Then

$$\begin{aligned}
R_n(F) &\leqslant R_n\left(\frac{1}{2}\left(g(K^d)+1\right)\right) \leqslant \sqrt{\frac{\pi}{2}}G_n\left(\frac{1}{2}\left(g(K^d)+1\right)\right) \\
&\leqslant \sqrt{\frac{\pi}{2}}\left(\frac{1}{2}G_n(g(K^d)) + \frac{1}{2\sqrt{n}}\right) \leqslant \sqrt{\frac{\pi}{2}}\left(dG_n(K) + \frac{1}{2\sqrt{n}}\right) \\
&\leqslant \sqrt{\frac{\pi}{2}}\left(2dG_n(H) + \frac{2d+1}{2\sqrt{n}}\right)
\end{aligned}$$

Note. But $G_n(H)$ is not easy to compute!

Experiments

Experiments

We want to see if the each bound is practical, i.e.

- there are minimizers in our classes of interest such that its value is less than 1,
- result coincides with train-test split.

We experiment with neural networks and kernel SVM.

Neural Networks - Adjusting Regularization Parameter

Putting Theorem 7 and 8 together and use $\hat{R}_n(F) \leqslant \sqrt{\frac{\pi}{2}}\hat{G}_n(F)$, we have

$$\mathbb{E}[\mathbf{1}(Y \neq f(X)] \leqslant \hat{\mathbb{E}}_n[\mathbf{1}(Y \neq f(X)] + \frac{LB(4\pi \ln k)^{1/2}}{n} \max_{j,j'} \sqrt{\sum_{i=1}^{n}(x_{ij} - x_{ij'})^2} + \sqrt{\frac{8\ln(2/\delta)}{n}}.$$

Choose $\delta = 0.05$.

Dataset: subset of Iris with Versicolor $(y = 0)$ and Virginica $(y = 1)$

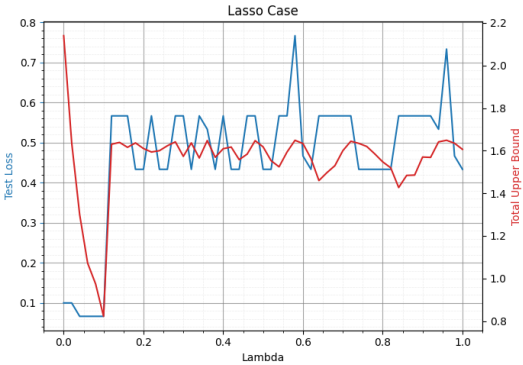
Model: 2-layer with activation *Tanh* (L=1).

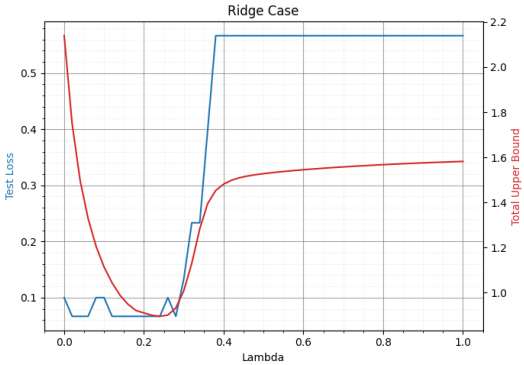
Losses (at the end we compute empirical $0 - 1$ loss)

$$\texttt{nn.BCEWithLogitsLoss()} + \lambda \sum_{w} |w| \ \text{(Lasso)}$$

$$\texttt{nn.BCEWithLogitsLoss()} + \lambda \sum_{w} |w|^2 \ \text{(Ridge)}$$

Neural Networks - Adjusting Regularization Parameter





Remark. The bound gives $\lambda = 0.1$ in Lasso case and $\lambda = 0.25$ in Ridge case, coinciding with the optimal case of test loss.

Neural Networks - Adjusting Depth

Proposition 7

Let $F_2 = \{\}$

Proposition 8

Define a class of D-layer networks with $D \geqslant 2$

$$F_D = \left\{ x \mapsto \sum_j w_j \sigma(h_j(x)) : \forall j, h_j \in F_{D-1} \text{ and } \|w\|_1 \leqslant B_D \right\},$$

where the activation σ is L-Lipschitz. Then $\hat{G}_n(F_D) \leqslant L B_D \hat{G}_n(F_{D-1})$

We want to demonstrate that adjusting depth using the bound is *not* robust.

We choose a larger dataset `Breast_Cancer`, fix a width and continually add layers.

Neural Networks - Adjusting Regularization Parameter

The average of $B_1, \ldots, B_D$ is *always* higher than 1.

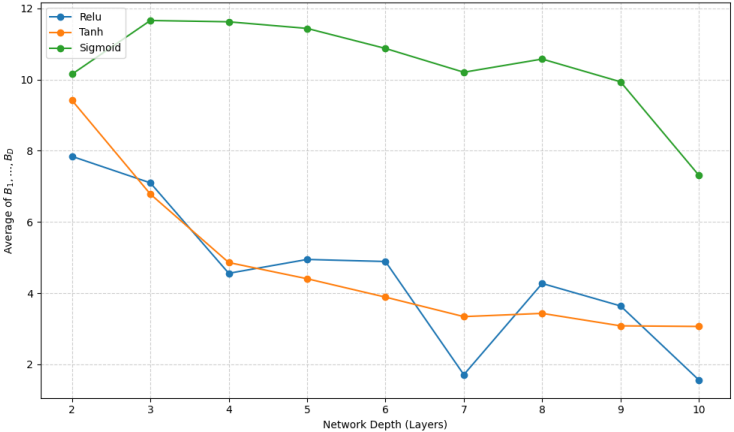
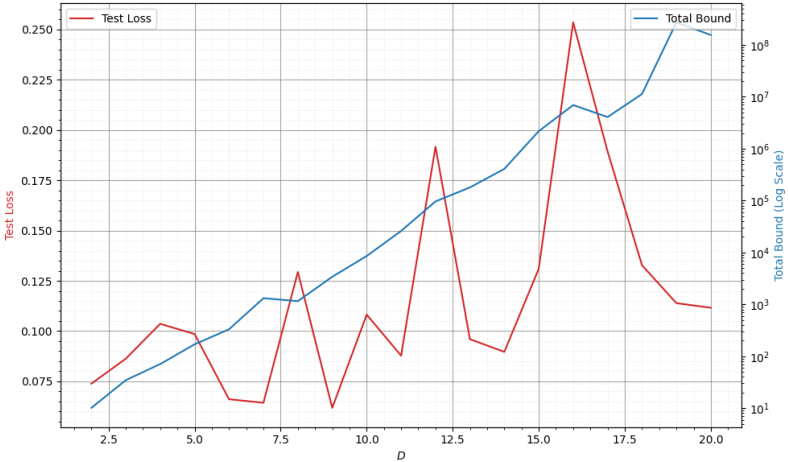


Figure: Average of $B_1, \ldots, B_D$ for different D's

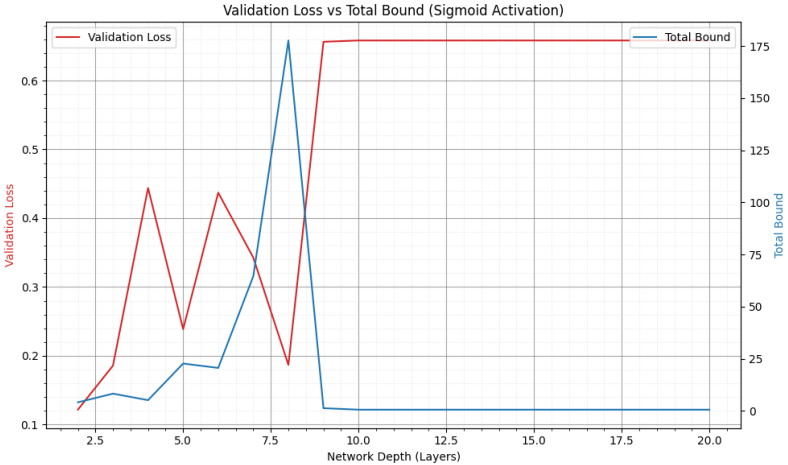
Neural Networks - Adjusting Depth

For *Tanh* and *ReLU* ($L = 1$), the bound explodes, but the models do not overfit. It fails to capture good models.



Neural Networks - Adjusting Depth

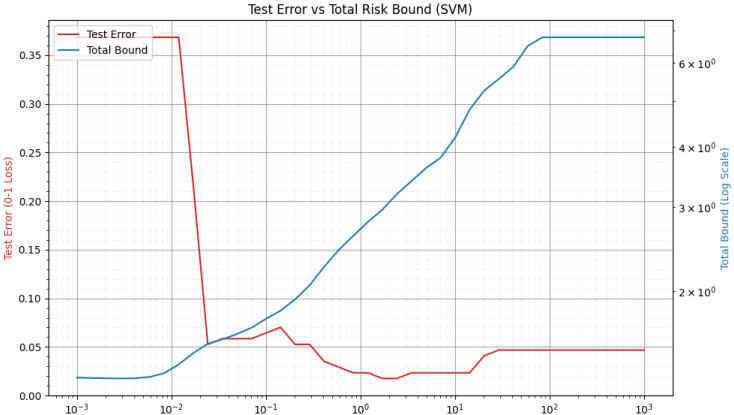
For *Sigmoid* ($L = 0.25$), even when the risk bound does not explode, it nominates models having vanishing gradients.



Kernel Methods - Adjusting Regularization Parameter C for SVM

Recall the kernelized objective

$$\max_{\alpha} \sum_{i=1}^{n} \alpha_i - \frac{1}{2} \sum_{i,j=1}^{n} \alpha_i \alpha_j y_i y_j K(x_i, x_j), \text{s.t.} \sum_{i=1}^{n} \alpha_i y_i = 0 \text{ and } 0 \leqslant \alpha_i \leqslant C.$$



Thank you for listening !